Daily Integral - Medium

Find the area between the curves:

$$f(x) = \frac{x^2 - 2x - 1}{x - 1} \;\&\; h(x) = \frac{x^2 + 8x - 20}{x}$$

between the points where they intersect.

This problem was generously submitted by Anzhey S. on https://dailyintegral.com.

Solution

First, we find the points of intersection, set $f(x) = h(x)$:

$$\begin{aligned}
&\frac{x^2 - 2x - 1}{x - 1} = \frac{x^2 + 8x - 20}{x} \\
&\Rightarrow x\left(x^2 - 2x - 1\right) = (x - 1)\left(x^2 + 8x - 20\right) \\
&\Rightarrow x^3 - 2x^2 - 1 = x^3 + 8x^2 - 20x - x^2 - 8x + 20 \\
&\Rightarrow -9x^2 + 27x - 20 = 0 \\
&\Rightarrow -(3x - 4)(3x - 5) = 0.
\end{aligned}$$

Hence, the points of intersection are at $x = 4/3$ and $x = 5/3$. Now, we simply $f(x)$ and $h(x)$:

$$\begin{aligned}
f(x) &= \frac{x^2 - 2x + 1 - 2}{x - 1} = \frac{(x - 1)^2 - 2}{x - 1} = x - 1 - \frac{2}{x - 1}, \\
h(x) &= \frac{x^2}{x} + \frac{8x}{x} - \frac{20}{x} = x + 8 - \frac{20}{x}.
\end{aligned}$$

We can find that on the interval $x \in [4/3, 5/3]$, $h(x) > f(x)$. Hence, the integral is:

$$\begin{aligned}
&\int_{4/3}^{5/3} [h(x) - f(x)]\, dx \\
=&\int_{4/3}^{5/3} \left[x + 8 - \frac{20}{x} - x + 1 + \frac{2}{x - 1}\right] dx \\
=&\int_{4/3}^{5/3} \left(9 - \frac{20}{x} + \frac{2}{x - 1}\right) dx \\
=&\ 9x - 20\ln(x) + 2\ln(x - 1)\Big|_{4/3}^{5/3} \\
=&\ 15 - 20\ln\left(\frac{5}{3}\right) + 2\ln\left(\frac{2}{3}\right) - 12 + 20\ln\left(\frac{4}{3}\right) - 2\ln\left(\frac{1}{3}\right) \\
=&\ 3 - 20\ln\left(\frac{5}{4}\right) + 2\ln(2).
\end{aligned}$$